

Advanced Public Economics — Applied Tutorial

Session VII: Tax Reforms

Daniel Weishaar

Roadmap for Session VII

- ① Recap lecture formulas on the **revenue and welfare implications of tax reforms**.
- ② Analyze (small) reforms of the status quo tax and transfer system in Germany.
 - 2.1 Pareto-improving reforms.
 - 2.2 Welfare-improving reforms.

Recap from Lecture — Revenue Functions

Under quasi-linear preferences and iso-elastic effort costs, revenue function

$$\mathcal{R}(y) = \underbrace{1 - F_y(y)}_{?} - \underbrace{\varepsilon y f_y(y) \frac{T'(y)}{1 - T'(y)}}_{?}$$

gives the **revenue impact** of a small **increase of marginal tax rates** by τ in a small income bracket starting at income level y .

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⇒ What about extensive margin responses?

Recap from Lecture — Revenue Functions — Extensive Margin (I)

The formula under intensive margin responses can be extended — requires some **fixed cost of working** in model in addition to variable effort costs.

$$\mathcal{R}_{int}(y) = \underbrace{1 - F_y(y)}_{\text{Mechanical Effect}} - \underbrace{\varepsilon_y f_y(y) \frac{T'(y)}{1 - T'(y)}}_{\text{Behavioral Response, Int. Margin}}$$

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Interpretation:

- ▶ An increase in marginal tax rates at income level y leads to a higher tax liability for everyone with income above.
- ▶ For some of these individuals, the increase in tax liability means that they are no longer willing to pay the fixed cost of working, they stop working completely.

Recap from Lecture — Revenue Functions — Extensive Margin (II)

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- $\pi_0(y')$: Participation elasticity, i.e., what share of individuals leave the labor market when net income is decreased by one percent. $\Rightarrow$ As for ε , we make assumptions regarding π_0 .

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- ▶ $\pi_0(y')$: Participation elasticity, i.e., what share of individuals leave the labor market when net income is decreased by one percent. $\Rightarrow$ As for ε , we make assumptions regarding π_0 .
- ▶ $t_p(y')$: Participation tax rate, i.e., change in net tax liability when individual starts to work. $\Rightarrow$ Can be estimated from effective tax schedule.

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We have already estimated most of the **ingredients** for the estimation of **revenue functions for Germany**:

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 - ▶ Singles without children ✓

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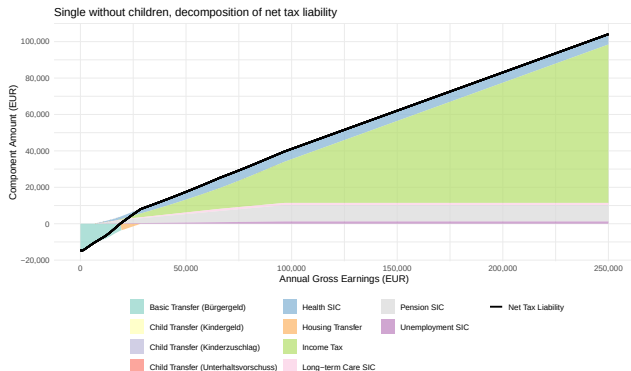
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⇒ *Note*: For single household with children, assume children of age 9 and age 5, rent of 650 EUR and 130 EUR heating costs. Other ingredients can be taken over from the single household without children.

Revenue Function Ingredients — Tax Liability and EMTR

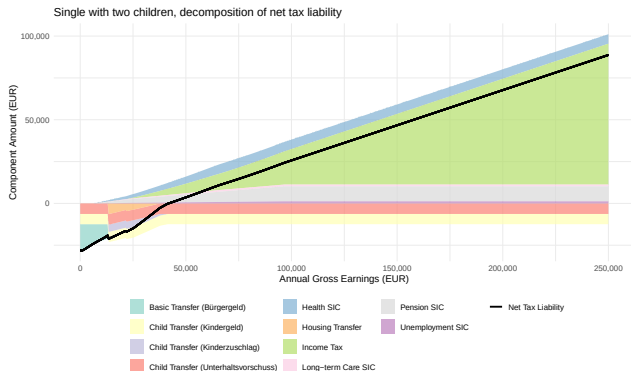
► Singles without children ✓

Singles without Children:



Revenue Function Ingredients — Tax Liability and EMTR

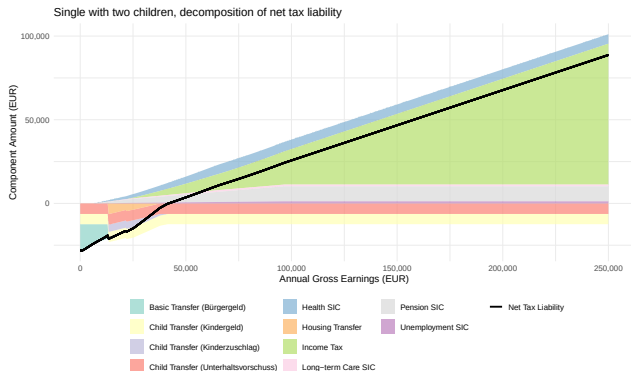
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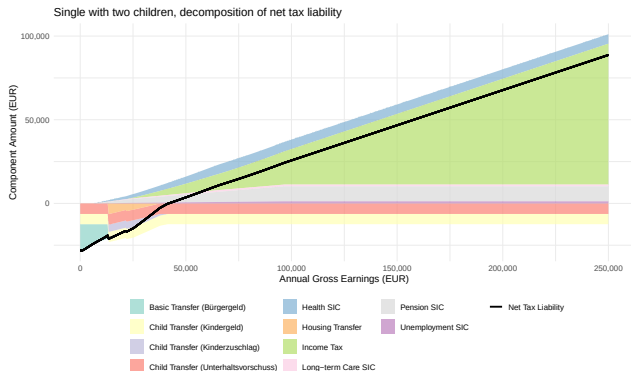
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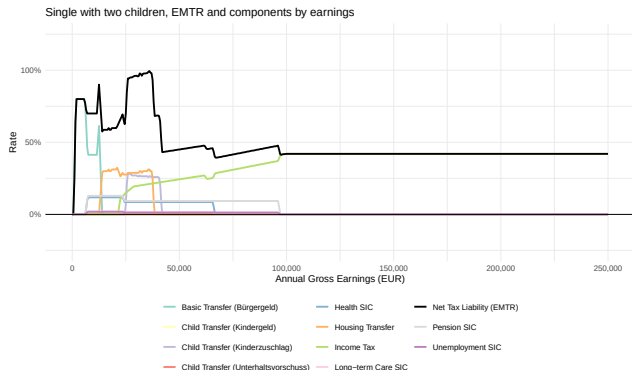
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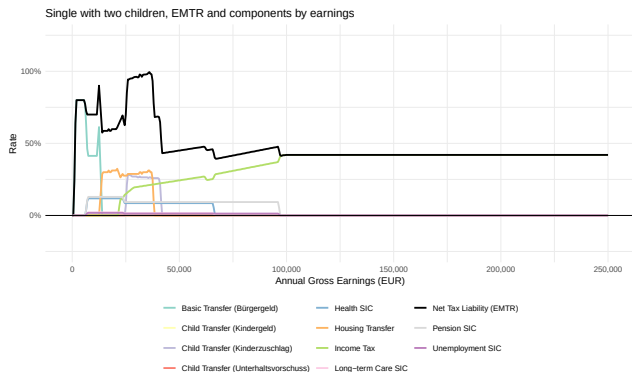
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- ▶ EMTR schedule features *kinks* — discrete changes in the marginal tax rate.

Revenue Function Ingredients — Tax Liability and EMTR

Singles with two Children:



Question from last time: who administers the different transfers? [Details](#)

- ▶ Singles without children ✓
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Revenue Function Ingredients — Participation Tax Rate

The extensive margin requires an estimate of the **participation tax rate** t_p . It measures, how much the net tax liability changes when a person starts working.

For $y > 0$, define the participation tax rate as

$$t_p(y) = \frac{T(y)}{y},$$

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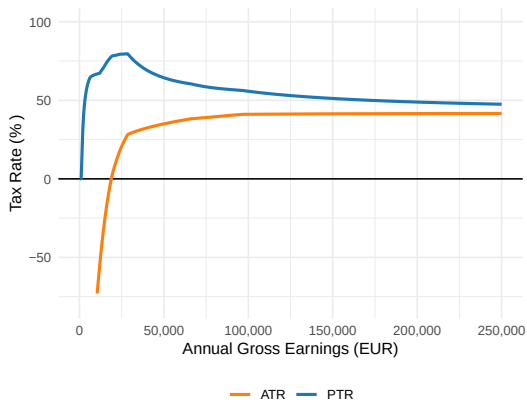
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⇒ **Hands-on exercise:** Calculate and plot the participation tax rate and average tax rate for singles with and without children in Germany.

Revenue Function Ingredients — Participation Tax Rate — Results

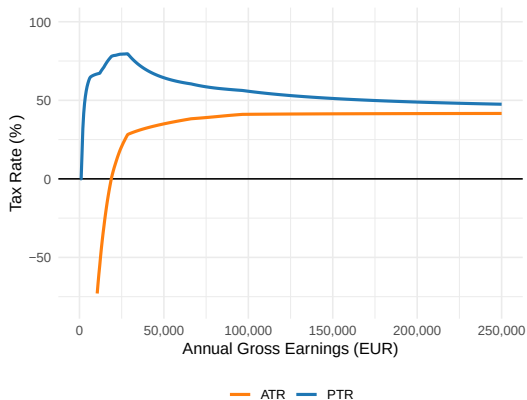
Singles without Children:



- Average tax rate (ATR) can be negative due to transfer, participation tax rate (PTR) non-negative.

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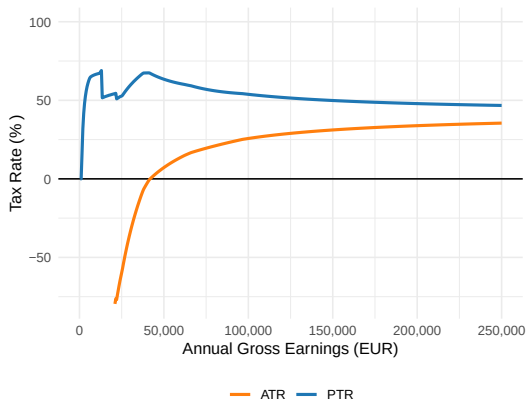
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- ▶ For singles with children, notches imply discrete jumps in ATR and PTR.

Revenue Function — Estimation

We now have all ingredients to compute revenue functions (incl. extensive margin).

$$\mathcal{R}_{int+ext}(y) = \underbrace{1 - F_y(y)}_{\text{Mechanical Effect}} - \underbrace{\varepsilon_y f_y(y) \frac{T'(y)}{1 - T'(y)}}_{\text{Behavioral Effect, Int. Margin}} - \underbrace{\int_y^\infty f_y(y') \pi_0(y') \frac{t_p(y')}{1 - t_p(y')} dy'}_{\text{Behavioral Effect, Ext. Margin}}$$

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► Effective marginal tax rates ✓

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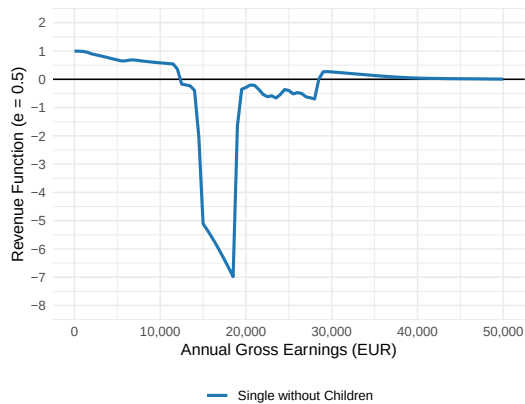
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⇒ **Hands-on exercise:** Estimate revenue functions, start with intensive margin only.

Revenue Function — Results — Intensive Margin Only (I)

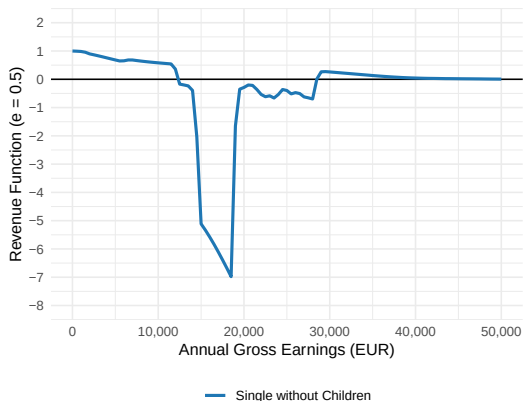
Singles without Children ($\varepsilon = 0.5$):



- Revenue function shows the revenue effect of a small MTR increase at a specific income level.

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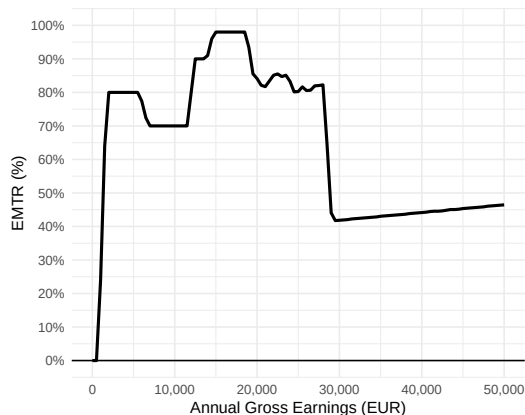
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- ▶ Revenue function shows the revenue effect of a small MTR increase at a specific income level.
- ▶ Bierbrauer et al. (2023b) provide necessary and sufficient conditions for Pareto efficiency related to revenue functions. [Details](#)
- ① **Negative values** indicate that raising tax rates comes with a loss in tax revenue. The system is inefficient between 12,000 EUR and 28,000 EUR, i.e., a one-bracket tax cut in this range realizes a Pareto improvement.

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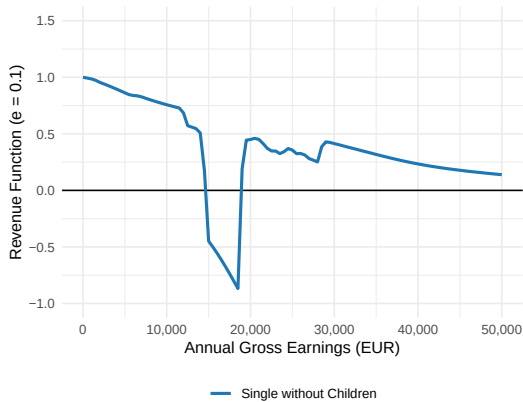
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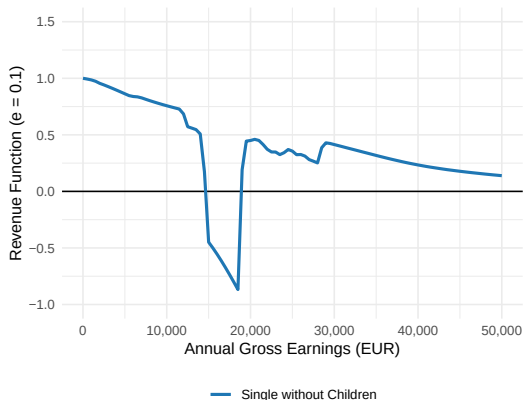
Revenue Function — Results — Intensive Margin Only (II)

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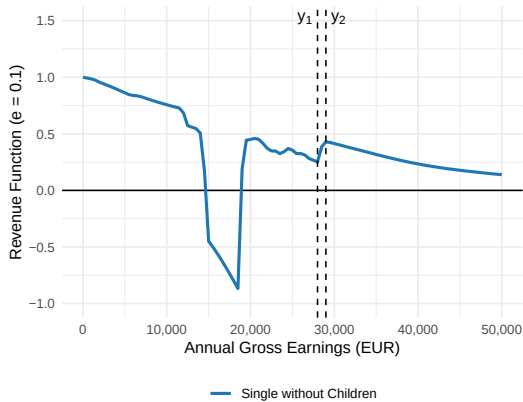
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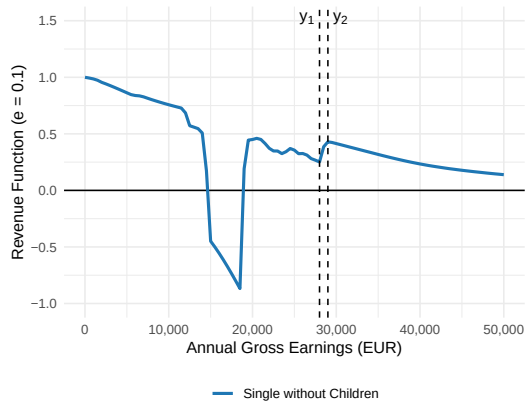


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- Compare the revenue effect at two income levels, y_1 to the left of the increase and y_2 , to the right.

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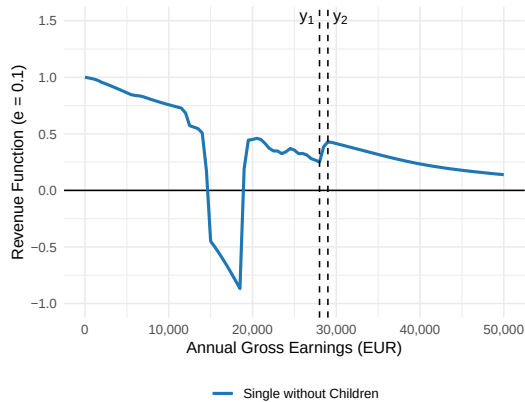


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- Compare the revenue effect at two income levels, y_1 to the left of the increase and y_2 , to the right.
- At y_1 , the mechanical effect is larger than for y_2 , because more are affected by the tax hike. Still, revenue effect is smaller at y_1 .

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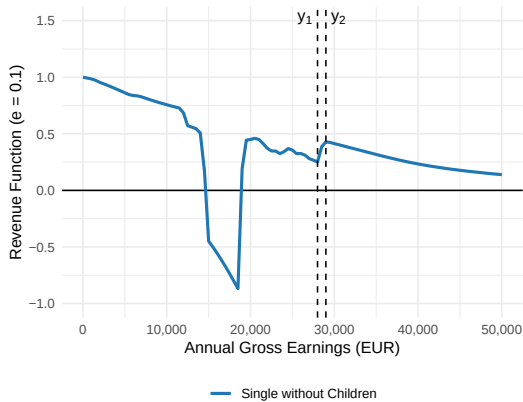


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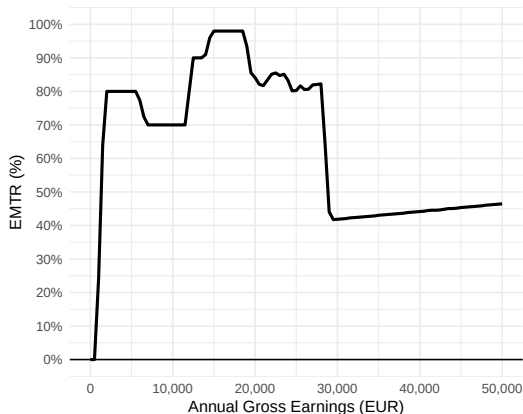


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- ▶ At y_1 , the mechanical effect is larger than for y_2 , because more are affected by the tax hike. Still, revenue effect is smaller at y_1 .
- ▶ Cure inefficiency by small decrease of MTR at y_1 and increase at y_2 .

Revenue Function — Results — Intensive Margin Only (II)

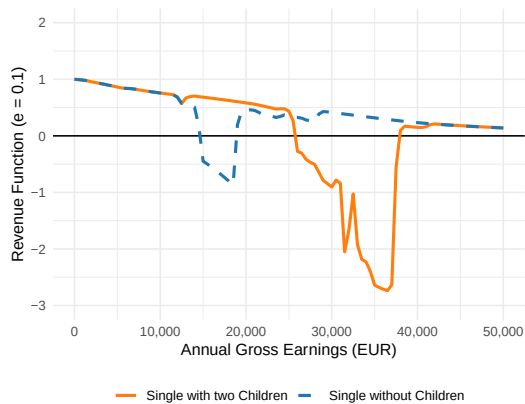
Singles without Children ($\varepsilon = 0.5$):



- ▶ Revenue function jumps observed where EMTR drops. Also relevant outside of Germany (Hansen and Weishaar, 2025). [Details](#)
- ▶ In Hansen and Weishaar (2025) we show that a *discrete drop* of MTR is a strong sign for an inefficiency for a broad class of different models.

Revenue Function — Results — Intensive Margin Only (III)

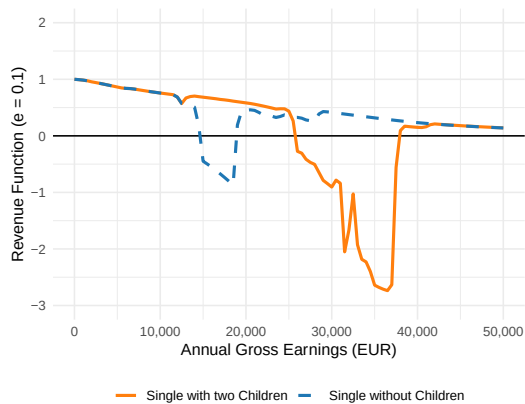
No Children vs. Children ($\varepsilon = 0.1$):



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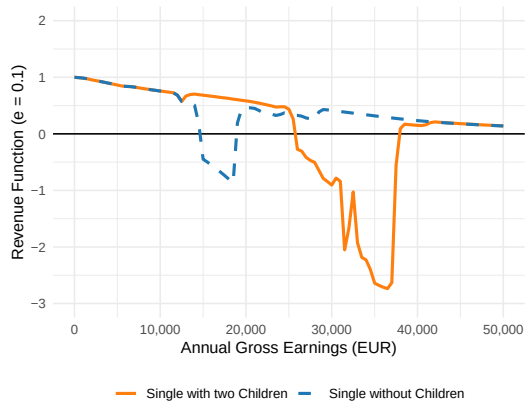
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- Inefficiency is more pronounced for singles with children.

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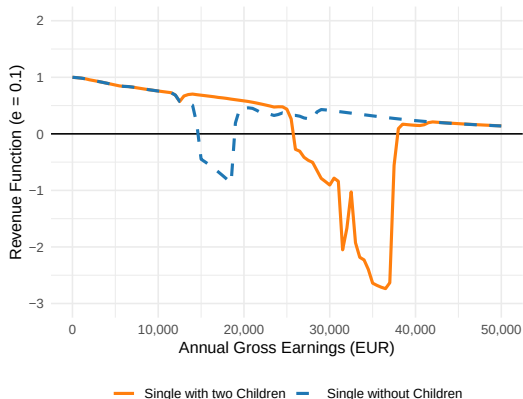


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- ▶ Inefficiency is more pronounced for singles with children.

Note of Caution: We have used here the general income distribution of singles. Ideally we would need to differentiate between singles with and without children and use the year corresponding to the tax system. We did not do so, since we only have access to SOEP training data.

Revenue Function — Results — Intensive Margin Only (III)

No Children vs. Children ($\varepsilon = 0.1$):



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Let's now include extensive margin responses. $\Rightarrow$ [Hands-on exercise](#)

Revenue Function — Results — Extensive Margin

$$- \int_y^\infty \underbrace{f_y(y') \pi_0(y') \frac{t_p(y')}{1 - t_p(y')}}_{\text{Ext. Margin (Inner Part)}} dy'$$

- Use numerical integration to obtain the extensive margin part.

Revenue Function — Results — Extensive Margin

Income	Ext. Margin (Inner)
0	0.05
500	0.01
1000	0.03
1500	0.1
...	0.2
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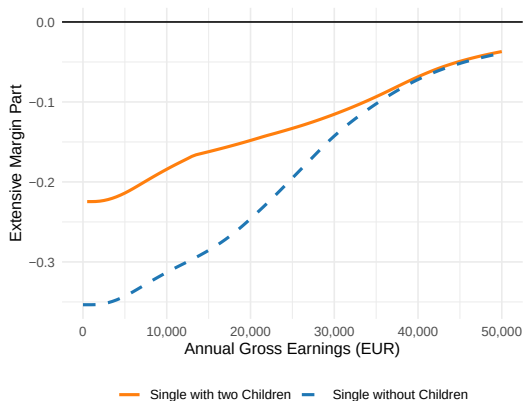
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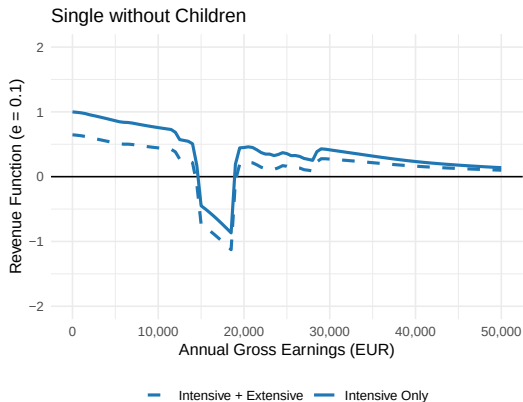
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- Use numerical integration to obtain the extensive margin part.
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- Extensive margin part shifts revenue function downward.

Revenue Function — Results — Extensive Margin



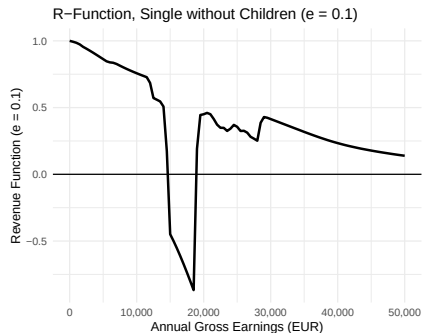
⇒ Under extensive margin responses, a tax hike has lower revenue effects, because it triggers some people to stop working.

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Reforming the System — Small Reform

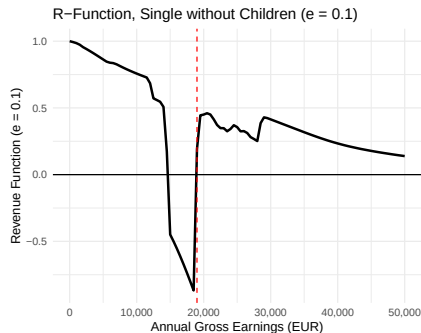
Revenue function shows inefficiencies and direction for reform. $\Rightarrow$ **Question:**
Where exactly should we change marginal tax rates?



- We know that an increasing revenue function indicates inefficiency.

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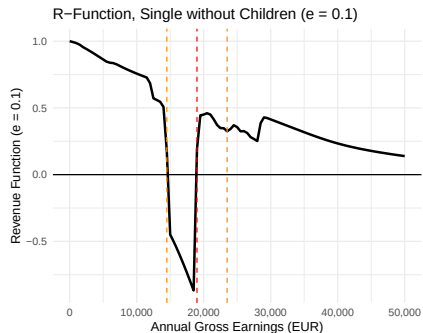
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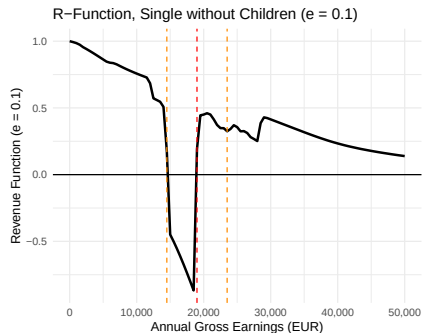
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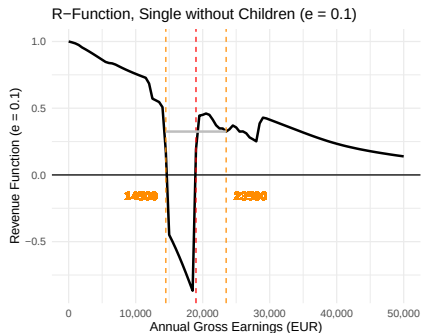
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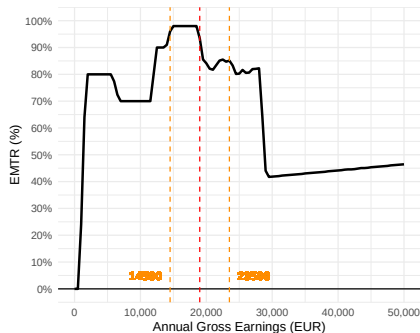


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Reforming the System — From Small Reform to Large Reforms (I)

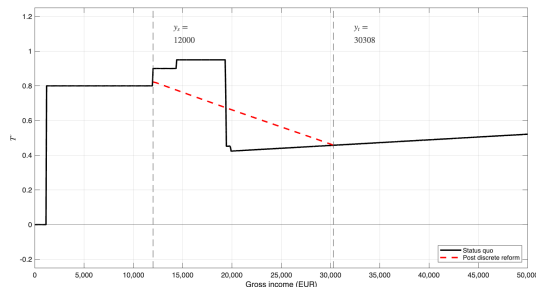
The preceding analysis gives us evidence about reform directions (**small reforms**). It does not answer the question, by how many percentage points we should change marginal tax rates (**large reforms**).

- ▶ Analyses of large reforms require a structural model, i.e., we would need to estimate the individual type distribution — as in session 03 for optimal tax rates.
- ▶ **Reason:** With large reforms, we need to describe the behavior of individuals *away from the status quo*. Not relevant for reform directions, because here we *stay close to status quo*.

Reforming the System — From Small Reform to Large Reforms (II)

⇒ We won't analyze large reforms here on our own. Ongoing work suggests that Pareto-improving large reforms that maximize tax revenue would lead to a **smoothly declining MTR schedule**:

Figure: Singles without Children, 2018



Source: Hansen and Weishaar (2025)

From Revenue Functions to Welfare Functions

Revenue functions describe the pure revenue effect of a local tax increase. A social planner might not always like to increase revenue through higher taxes, because she cares about those people from which these taxes are levied. $\Rightarrow$ **Welfare Analysis**.

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Welfare Effect of Reforms (here under intensive margin responses):

$$\begin{aligned}\mathcal{W}(y) &= \mathcal{R}(y) - (1 - F_y(y)) \mathcal{G}(y) \\ &= (1 - F_y(y)) - \varepsilon_y y f_y(y) \frac{T'(y)}{1 - T'(y)} - (1 - F_y(y)) \mathcal{G}(y)\end{aligned}$$

where $\mathcal{G}(y)$ is the average welfare weight of individuals with incomes larger than y .

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where $\mathcal{G}(y)$ is the average welfare weight of individuals with incomes larger than y .

Special Cases:

- ▶ Under equal weights, i.e., $g(y) = 1$, and for $T' > 0$: $\mathcal{W}(y) < 0$.
- ▶ Under Rawlsian weights, i.e., $g(y) = 0$ for $y > 0$: $\mathcal{W}(y) = \mathcal{R}(y)$.

Welfare Functions — Estimation

Some **Hands-on exercises**: Estimate the welfare effect of local tax increases for singles without children under the following hypothetical preferences different social planners:

- ① **Working Poor**: concentrates all weight on singles with positive incomes of less than 30,000 EUR
- ② **Inequality Averse**: assigns declining weights in a way that a one percent increase in the income level decreases weights by 0.25 percent.
- ③ **Loves the Rich**: assigns linearly increasing weights.

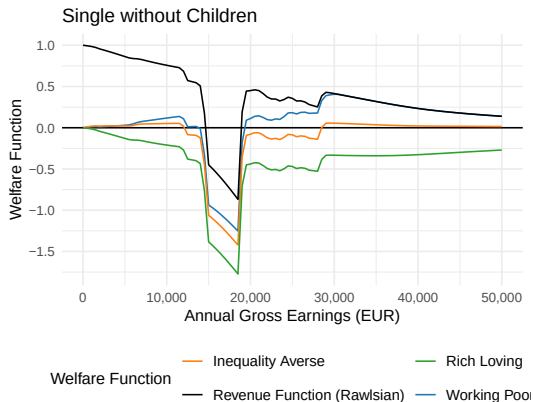
Note: Normalize all welfare weights to have a mean of 1.

Welfare Functions — Results



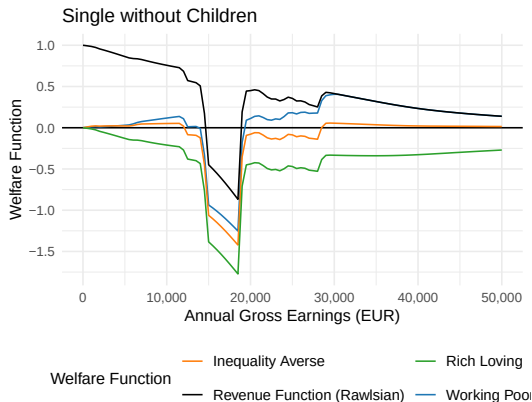
- $G(y)$ can take different forms, all $G(0) = 1$.

Welfare Functions — Results



- ▶ $\mathcal{G}(y)$ can take different forms, all $\mathcal{G}(0) = 1$.
- ▶ Welfare functions can agree or disagree to a tax hike at a specific income level y .

Welfare Functions — Results



- ▶ $\mathcal{G}(y)$ can take different forms, all $\mathcal{G}(0) = 1$.
- ▶ Welfare functions can agree or disagree to a tax hike at a specific income level y .
- ▶ Welfare functions always agree when tax system is inefficient (for non-negative welfare weights).

Current Research Frontier

Current Research Frontier — Overview

Current research works on novel tools and extensions for the analysis of tax reforms:

- ① Analysis of **large reforms** (Hansen and Weishaar, 2025).
- ② Analysis of **reforms under misperceptions** (Hansen and Weishaar, 2025).
- ③ Analysis of **reforms with multiple dimensions** (Bierbrauer et al., 2023a; Bierbrauer et al., 2024).
- ④ Analysis of reforms with **endogenous take-up** (Bastani et al., 2026; Leroy, 2024).

Supplementary Material — Tax Reforms under Misperceptions (I)

Background: Discussed models assume rational behavior and perfect information about tax and transfer system. However, empirical research shows that this is not plausible, i.e., people have misperceptions about the tax code.

Question: Do inefficiencies in the tax code persist, if people have misperceptions, e.g., if they take their average tax rates to be the marginal one (*ironing*)?

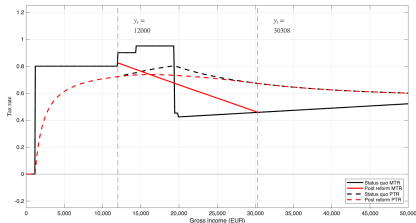
⇒ Brief look into Hansen and Weishaar (2025).

Method:

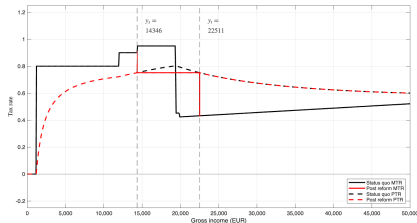
- ▶ Develop new tests for Pareto efficiencies under misperceptions.
- ▶ Derive revenue function formulas under misperceptions and estimate them empirically for a large set of countries.
- ▶ Calibrate revenue maximizing small and large tax reform.

Supplementary Material — Tax Reforms under Misperceptions (II)

→ **Example: Germany** — Single individual, no children, legal framework of 2018.



(a) Rational



(b) Non-Rational Ironing

⇒ Large reform under misperceptions (ironing) does not lead to complete smoothing of tax rates, but introduces an intermediate bracket.

Supplementary Material — Tax Reforms with Multiple Dimensions (I)

Background: Discussed models assume one input to the tax function and a single dimension of behavioral responses. In reality, tax system contains multiple inputs (as we saw in the microsimulation part).

Question: How can we analyze the efficiency and welfare consequences of tax reforms when we have multiple inputs to the tax function and correspondingly multiple behavioral responses?

⇒ Brief look into Bierbrauer et al. (2024).

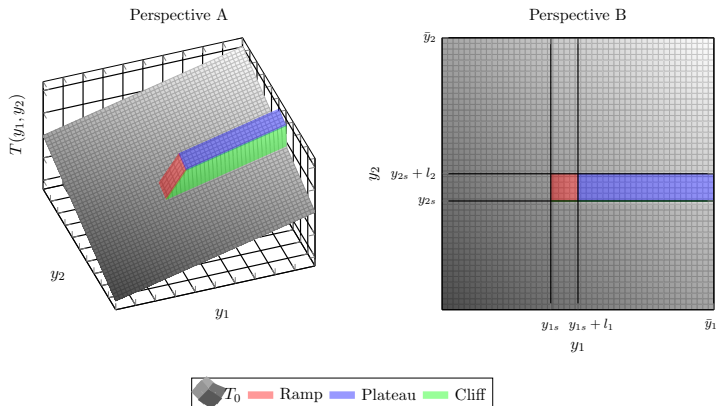
Method:

- ▶ Develop tests for Pareto efficiency under multidimensional heterogeneity.
- ▶ Derive revenue function formulas in one setting with multidimensional heterogeneity: the taxation of couples.
- ▶ Calibrate (conditional) revenue functions for the United States and estimate efficiency and welfare consequences.

Supplementary Material — Tax Reforms with Multiple Dimensions (II)

There are now two fundamental types of reforms:

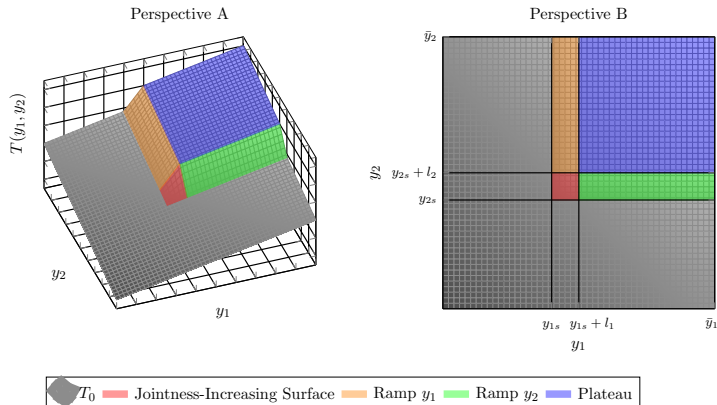
- ① **Change MTR** on primary earnings. **conditional** on secondary earnings.



Supplementary Material — Tax Reforms with Multiple Dimensions (II)

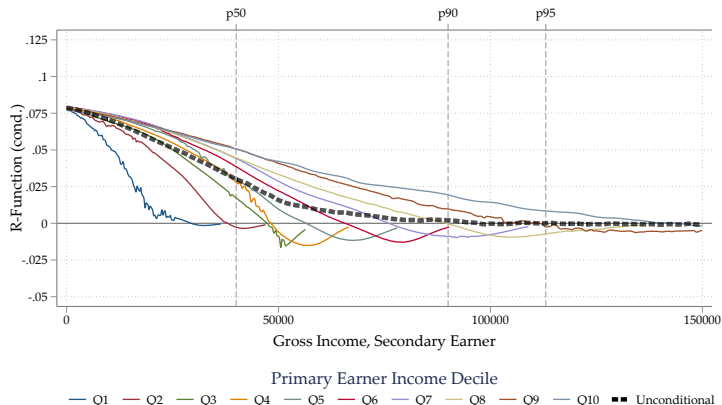
There are now two fundamental types of reforms:

- ① **Change MTR** on primary earnings. **conditional** on secondary earnings.
- ② Modify the **jointness** of the tax system, i.e., how MTR on primary earnings change with an increase of secondary earnings.



Supplementary Material — Tax Reforms with Multiple Dimensions (III)

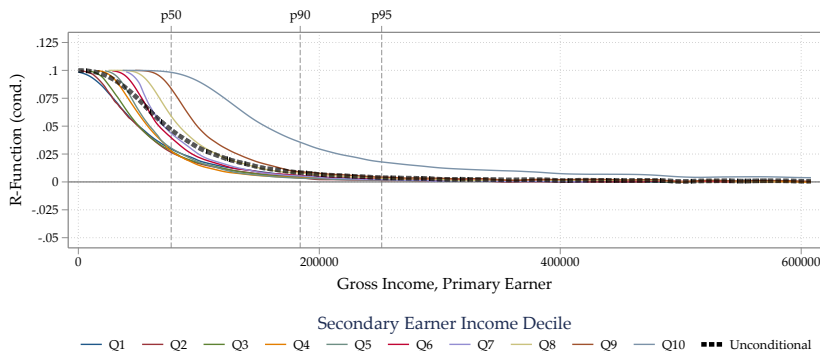
Changing Secondary Earner MTR conditional on Primary Earnings



- ▶ There are inefficiencies in the tax treatment of secondary earnings.
- ▶ Curing inefficiencies require targeted reforms conditional on primary earnings.

Supplementary Material — Tax Reforms with Multiple Dimensions (III)

Changing Primary Earner MTR conditional on Secondary Earnings



- There are no such inefficiencies in the tax treatment of primary earnings.

That's it! It was a pleasure!

References

References I

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-  LEROY, C. (2024). *Raising Take-up of Welfare Programs: Evidence from a Large French Reform*. Tech. rep. Working paper.

Bierbrauer et al. (2023b) — Sufficient Conditions

Bierbrauer et al. (2023b) characterize Pareto efficiency in relation to revenue functions.

Theorem (Bierbrauer et al. (2023))

There is a Pareto-improving tax reform unless

- ▶ $\mathcal{R}_0(y) \in [0, 1]$ for all $y \geq 0$ and
 - ▶ $\mathcal{R}_0$ is monotonically decreasing for all $y \geq 0$.
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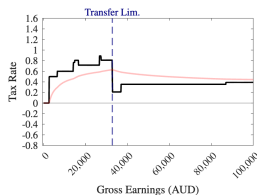
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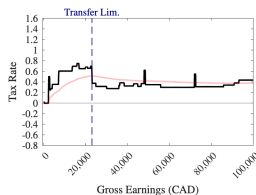
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 - ▶ $(\mathcal{R}_0(y) > 1 \rightarrow$ self-financing **one bracket tax increase** at y .)

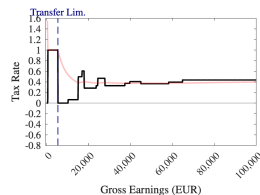
Hansen and Weishaar (2025) — Cross-Country Analysis — EMTR



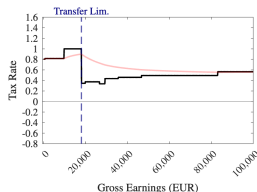
(a) Australia



(b) Canada



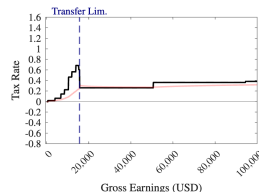
(c) Spain



(d) Finland



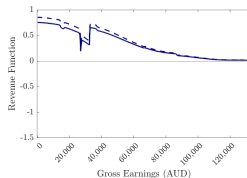
(e) UK



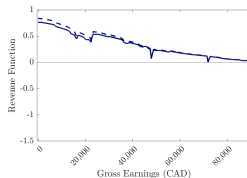
(f) USA

A lot of countries feature discrete MTR drops where transfers are phased out.

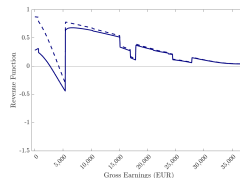
Hansen and Weishaar (2025) — Cross-Country Analysis — $\mathcal{R}$ -Function



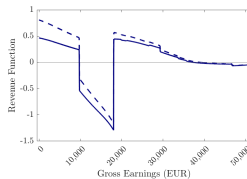
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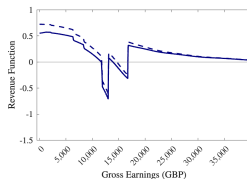
(b) Canada



(c) Spain



(d) Finland



(e) UK



(f) USA

Non-monotonicities indicate inefficiencies in all of the selected countries.

Who administers the different transfers?

Transfer	Executing Body	Executing Level	Main Ministry
Arbeitslosengeld I	Bundesagentur für Arbeit	Bund	Bundesministerium für Arbeit und Soziales
Bürgergeld (SGB II)	Jobcenter (Bundesagentur für Arbeit + Kommune)	Bund / Kommune	Bundesministerium für Arbeit und Soziales
Sozialhilfe (SGB XII)	Kommunale Sozialämter	Kommune	Bundesministerium für Arbeit und Soziales
Grundsicherung im Alter / bei Erwerbsminderung	Kommunale Sozialämter	Kommune	Bundesministerium für Arbeit und Soziales
Wohngeld	Kommunale Wohngeldstellen	Kommune	Bundesministerium für Wohnen, Stadtentwicklung und Bauwesen
Kindergeld	Familienkasse (bei der Bundesagentur für Arbeit)	Bund	Bundesministerium der Finanzen
Kinderzuschlag	Familienkasse (bei der Bundesagentur für Arbeit)	Bund	Bundesministerium für Familie, Senioren, Frauen und Jugend (fachlich), Bundesministerium der Finanzen (steuerlich)
Elterngeld	Elterngeldstellen der Länder	Länder	Bundesministerium für Familie, Senioren, Frauen und Jugend
Unterhaltsvorschuss	Jugendämter	Kommune	Bundesministerium für Familie, Senioren, Frauen und Jugend

See also interesting [FAZ article](#) from today about the plan to harmonize different transfers.